

Grade 8
Unit 4
Lesson 7 Practice

Name _____
Date _____
Period _____

Apply the laws of exponents to generate an equivalent algebraic expression in questions 1-6.

1. $x^7 \cdot x^4$

2. $7y \cdot 6y^{10}$

3. $(c^3)^5$

4. $(8z^2)^3$

5. $4a^4b^6 \cdot 2a^3b^2$

6. $(d^5 \cdot 6c^3)^2$

7. Match the algebraic expressions from Column A to an equivalent algebraic expression in Column B.

Column A	Column B
A. $(xy)^0 \cdot (5x^3 \cdot 2x^7)$	E. $7x^7y^7$
B. $((xy)^2)^3 \cdot 7xy$	F. $36x^{14}$
C. $(6x^7)^2$	G. $10x^{10}$
D. $(x^2y)^4 \cdot (4x)^3$	H. $64x^{11}y^4$

A rectangle has a length represented by the expression $2c^5$ and a width represented by the expression $7c^3$.

8. Draw a rectangle and label the dimensions.
9. The formula $\text{Area} = \text{length} \times \text{width}$ can be used to find the area of the rectangle. Using the formula, generate an algebraic expression to represent the area of the rectangle.
10. Simplify the expression from question 9 to find an equivalent expression that represents the area of the rectangle.